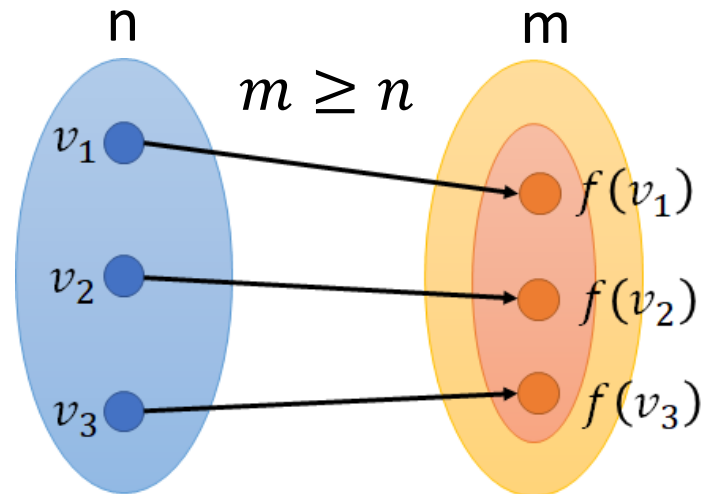


期中考總複習

常見題型: Linear Function

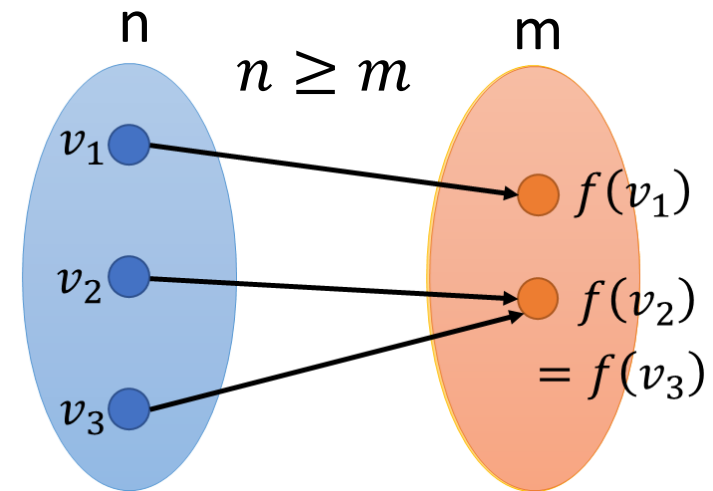
Linear Function

- Determining a function is linear or not.
- Determining a function is onto or one-to-one.



one-to-one

The columns of A are independent.



onto

Rank $A = m$

$A: m \times n$

4. In the following questions, T_1 , T_2 , and T_3 are linear transformations. (2014)

(a) (5%) Let $T_1 : \mathcal{R}^n \rightarrow \mathcal{R}^m$. Suppose $T_1(\mathbf{e}_i) = \mathbf{v}_i$, $i = 1, 2, \dots, n$ where $\mathbf{e}_i$ is the i th standard vector of $\mathcal{R}^n$. Find the standard matrix of T_1 .

(b) (10%) Let $T_2 : \mathcal{R}^n \rightarrow \mathcal{R}^m$. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis for $\mathcal{R}^n$ and suppose $T_2(\mathbf{u}_i) = \mathbf{w}_i$, $i = 1, 2, \dots, n$. Find the standard matrix of T_2 in terms of $\mathbf{u}_i$ and $\mathbf{w}_i$.

(c) (5%) Let $T_3 : \mathcal{R}^2 \rightarrow \mathcal{R}^3$. Suppose $T_3\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ and $T_3\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$. Find the standard matrix of T_3 .

(d) (5%) Is T_3 onto? Why? No

(e) (5%) Is T_3 one-to-one? Why? Yes

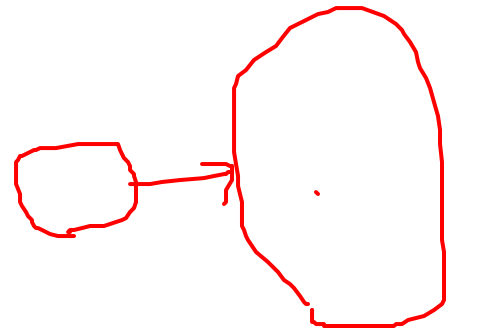
(a) $[v_1 \quad \dots \quad v_n]$

(b) $A[u_1 \quad \dots \quad u_n] = [w_1 \quad \dots \quad w_n]$

$A = [w_1 \quad \dots \quad w_n][u_1 \quad \dots \quad u_n]^{-1}$

(c) $T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = T\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$

$= T\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) - 2T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -3 \\ -4 \\ 1 \end{bmatrix}$



$\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \begin{bmatrix} -3 \\ -4 \\ 1 \end{bmatrix}$

5. (20%) Consider the following linear transformations:

(2015)

$$T_1 : \mathcal{R}^2 \rightarrow \mathcal{R}^3 \text{ defined by } T_1 \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + x_2 \\ x_1 - 3x_2 \\ 4x_1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ 1 & -3 \\ 4 & 0 \end{bmatrix}$$

$$T_2 : \mathcal{R}^3 \rightarrow \mathcal{R}^2 \text{ defined by } T_2 \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 - x_2 + 4x_3 \\ x_1 + 3x_2 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 4 \\ 1 & 3 & 0 \end{bmatrix}$$

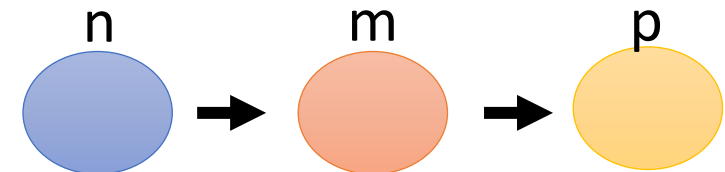
(a) (8%) Find the standard matrices of T_1 , T_2 , T_1T_2 and T_2T_1 .

(b) (4%) Is T_1T_2 onto? **No** Is T_1T_2 one-to-one? **No**

(c) (4%) Let $U_1 : \mathcal{R}^n \rightarrow \mathcal{R}^m$ and $U_2 : \mathcal{R}^m \rightarrow \mathcal{R}^p$ be linear. Determine if the following statements are true or false (explain your answers):

(i) If $m > n$, then U_2U_1 cannot be onto. **False**

(ii) If $m < p$, then U_2U_1 cannot be onto. **True**



(d) (4%) Let U_1 and U_2 be the linear transformations defined in (c). Prove that if U_1 and U_2 are one-to-one, then U_2U_1 is one-to-one.

Assume U_2U_1 is not one-to-one

$$U_2U_1(\mathbf{v}) = U_2U_1(\mathbf{u})$$

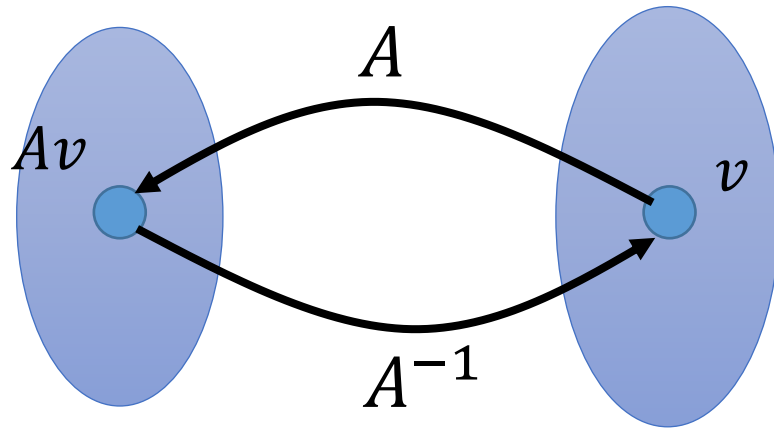
$$U_2(\mathbf{v}') = U_2(\mathbf{u}') \quad U_2 \text{ is not one-to-one}$$

Contradiction

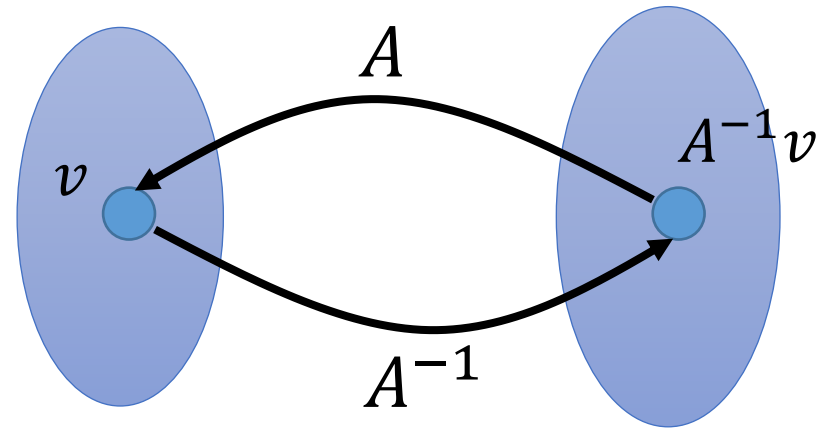
常見題型: Matrix Inverse

Invertible

- A is called invertible if there is a matrix B such that $AB = I$ and $BA = I$ ($B = A^{-1}$)



A must be one-to-one

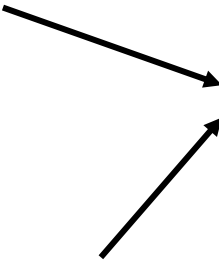


A must be onto

(不然 A^{-1} 的 input 就會有限制)

Invertible

- Let A be an $n \times n$ matrix.
 - Onto $\rightarrow$ One-to-one $\rightarrow$ invertible
 - The columns of A span $\mathbb{R}^n$
 - For every b in $\mathbb{R}^n$, the system $Ax=b$ is consistent
 - The rank of A is the number of rows
 - One-to-one $\rightarrow$ Onto $\rightarrow$ invertible
 - The columns of A are linear independent
 - The rank of A is the number of columns
 - The nullity of A is zero
 - The only solution to $Ax=0$ is the zero vector
 - The reduced row echelon form of A is I_n



Rank $A = n$

Invertible

An $n \times n$ matrix A is invertible.



The reduced row echelon form of A is I_n



A is a product of elementary matrices

$$R = \text{RREF}(A) = I_n$$

$$E_k \cdots E_2 E_1 A = I_n$$

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} I_n$$

$$= E_1^{-1} E_2^{-1} \cdots E_k^{-1}$$

Inverse for matrix product

- A and B are invertible nxn matrices, is AB invertible? yes

$$(AB)^{-1} = B^{-1}A^{-1}$$

- Let $A_1, A_2, \dots, A_k$ be nxn invertible matrices. The product $A_1A_2 \cdots A_k$ is invertible, and

$$(A_1A_2 \cdots A_k)^{-1} = (A_k)^{-1}(A_{k-1})^{-1} \cdots (A_1)^{-1}$$

- If A is invertible, is A^T invertible? $(A^T)^{-1} = ? \quad (A^{-1})^T$

Matrix Inverse

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \quad A + B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

2. (10%) Find the inverse of the following matrix.

會不會算 A^{-1}

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 8 \\ 2 & 5 & 6 \end{bmatrix} \quad (2018)$$

5. (15%) Let A be B be both invertible matrices of size $n \times n$. Answer whether the following statements are true or false. Justify your answer.

(a) (5%) $A + B$ is also invertible.

(b) (5%) AB is also invertible.

(c) (5%) $\det(B) \cdot A^{-1}$ is also invertible.

Inverse 的各種性質 (2016)

6. (15%) Consider the 2×4 matrix $A = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{bmatrix}$. (2016)

(a) (5%) Let R be the reduced row echelon form of A . Find R . (b) (5%) Find an invertible matrix P such that $PA = R$.

Inverse v.s. Row operation

$$E_k \cdots E_2 E_1 A = R$$

$$PA = R$$

Dependent/Independent

3. (a) (6%) Let Q be an $n \times n$ invertible matrix. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a linearly independent set of vectors in $\mathcal{R}^n$. Prove that $\{Q\mathbf{u}_1, Q\mathbf{u}_2, \dots, Q\mathbf{u}_k\}$ is linearly independent.

(2015)

$\{u_1, u_2, \dots, u_k\}$ independent

 $\{Qu_1, Qu_2, \dots, Qu_k\}$ independent

$$c_1 Qu_1 + c_2 Qu_2 + \dots + c_k Qu_k = 0$$

 $Q(c_1 u_1 + c_2 u_2 + \dots + c_k u_k) = 0$

Q^{-1} Q^{-1}

 $c_1 u_1 + c_2 u_2 + \dots + c_k u_k = 0$  $c_1 = c_2 = \dots = c_k = 0$

$\{u_1, u_2, \dots, u_k\}$ independent

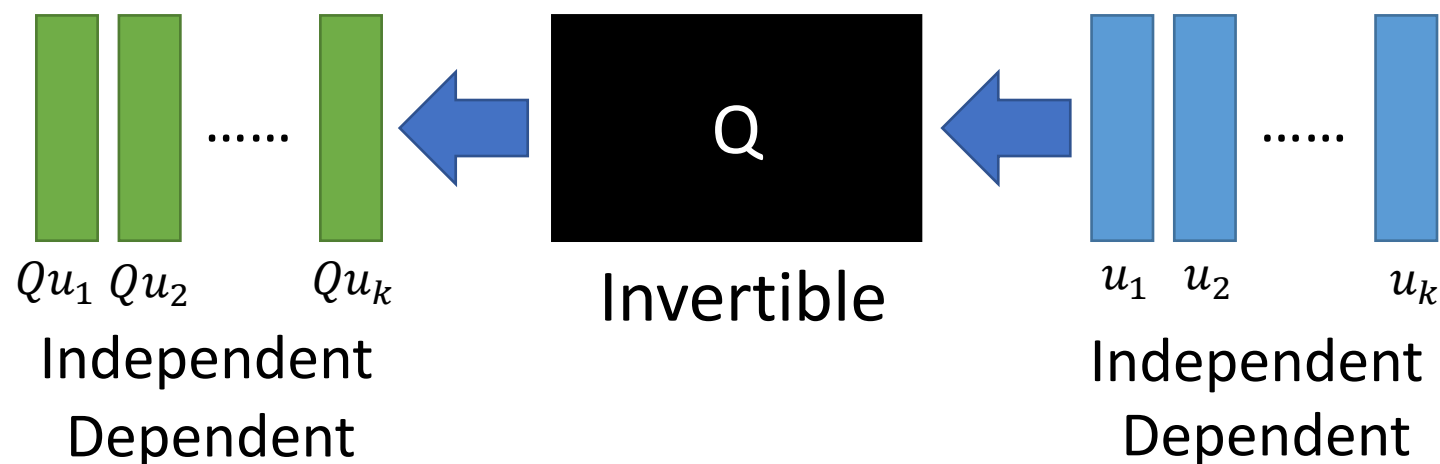
$\{u_1, u_2, \dots, u_k\}$ dependent

 $\{Qu_1, Qu_2, \dots, Qu_k\}$ dependent

Dependent/Independent

3. (a) (6%) Let Q be an $n \times n$ invertible matrix. Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a linearly independent set of vectors in $\mathcal{R}^n$. Prove that $\{Q\mathbf{u}_1, Q\mathbf{u}_2, \dots, Q\mathbf{u}_k\}$ is linearly independent.

(2015)



7. (15%) Let A be an $m \times n$ matrix and R be its reduced row echelon form. We know that there is an $m \times m$ invertible P such that $PA = R$.

(2015)

- (a) (8%) Use the formula $PA = R$ to prove that the rows of R are linearly independent if and only if the rows of A are linearly independent.

常見題型: Subspace & Basis & Dimension

Subspace

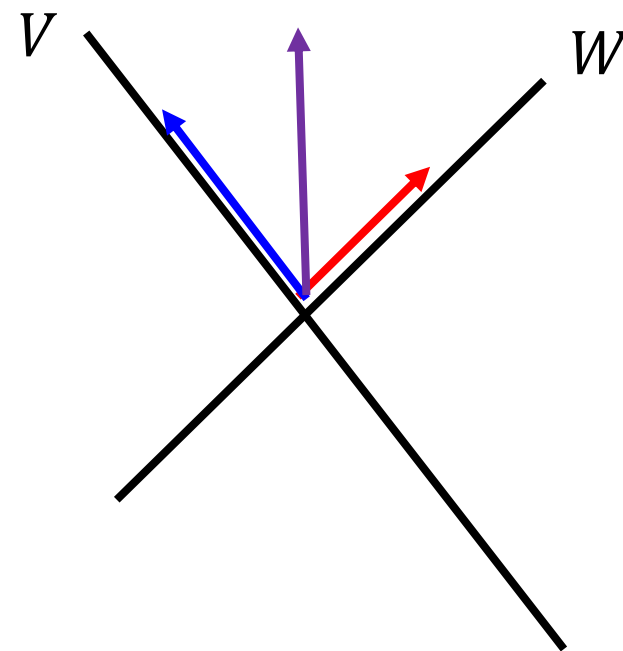
- 一定要會: 檢查一個 vector set 是不是 subspace

4. (10%) Let V and W be subspaces of R^n . Let $V + W = \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in V \text{ and } \mathbf{y} \in W\}$. Prove that $V + W$ is a subspace of R^n .

(2016)

注意:這不是聯集!

- 1 $\mathbf{0} \in V, \mathbf{0} \in W \Rightarrow \mathbf{0} \in V + W$
- 2 $\mathbf{x} \in V + W \Rightarrow \mathbf{x} = \mathbf{v} + \mathbf{w}$
 $\Rightarrow c\mathbf{x} = c\mathbf{v} + c\mathbf{w} \quad c\mathbf{v} \in V, c\mathbf{w} \in W$
 $\Rightarrow c\mathbf{x} \in V + W$
- 3 $\mathbf{x}_1 \in V + W, \mathbf{x}_2 \in V + W \dots\dots$



1. (15%) Let V and W be subspaces of $\mathcal{R}^n$.

(2020)

(a) (7%) If $V \cap W$ is a subspace of $\mathcal{R}^n$, then prove it. Otherwise, give a counterexample.

(b) (8%) Let $\mathbf{v}$ be a vector in V . Let $\mathbf{v} + W = \{\mathbf{v} + \mathbf{w} \mid \mathbf{w} \in W\}$. If $\mathbf{v} + W$ is a subspace of $\mathcal{R}^n$, then prove it. Otherwise, give a counterexample.

(a) 1 $\mathbf{0} \in V \cap W$

2 $x \in V \cap W$ $cx \in V \cap W?$

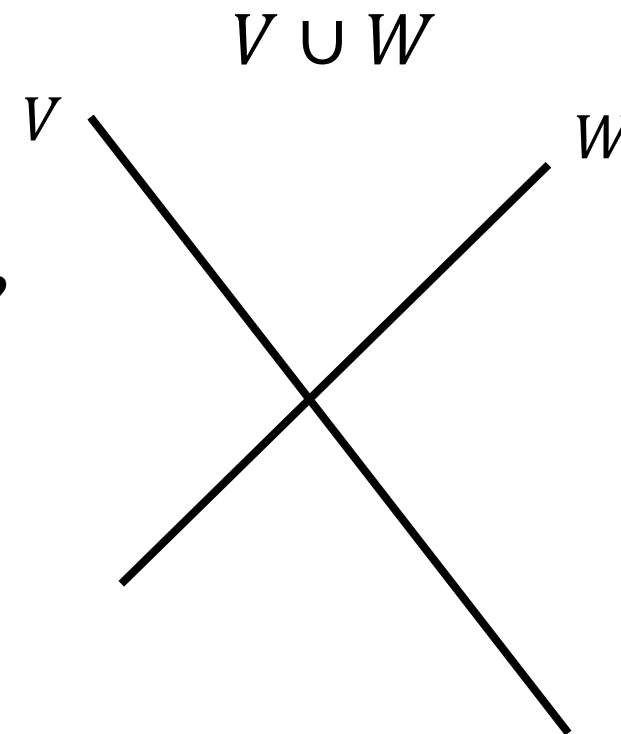
$x \in V, x \in W$ $cx \in V, cx \in W$

3 $x_1 \in V \cap W$ $x_2 \in V \cap W$ $x_1 + x_2 \in V \cap W?$

$x_1 \in V$ $x_2 \in V$ $x_1 + x_2 \in V$

$x_1 \in W$ $x_2 \in W$ $x_1 + x_2 \in W$

(b)



(2020)

8. (18%) Prove or disprove that the following subsets of $\mathcal{R}^n$ is a subspace. If it is a subspace, find the dimension of it.

(a) (6%) $\mathcal{S}_1 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathcal{R}^2 \mid x + y \geq 0 \right\} \subseteq \mathcal{R}^2.$ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $-1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(b) (6%) $\mathcal{S}_2 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathcal{R}^3 \mid (x + y + z - 1)^2 = 0 \right\} \subseteq \mathcal{R}^3$ No zero vector

(c) (6%) $\mathcal{S}_3 = \{A\mathbf{v} + x\mathbf{b} \mid \mathbf{v} \in \mathcal{R}^2, x \in \mathcal{R}\}$ where $A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \\ 10 & 11 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix}.$

$span \left\{ \begin{bmatrix} 1 \\ 4 \\ 7 \\ 10 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 8 \\ 11 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix} \right\}$

Basis

Why nonzero?

- Let V be a nonzero subspace of $\mathbb{R}^n$. A **basis** B for V is a **linearly independent generation set** of V .

$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis for $\mathbb{R}^n$.

- $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is independent
- $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ generates $\mathbb{R}^n$.

$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$

$\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$ $\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$ $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ any two independent vectors form a basis for $\mathbb{R}^2$

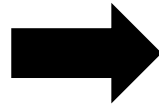
Theorem

- 1. A basis is the **smallest** generation set.
- 2. A basis is the **largest** independent vector set in the subspace.
- 3. Any two bases for a subspace **contain the same number of vectors**.
 - The number of vectors in a basis for a nonzero subspace V is called **dimension** of V ($\dim V$).

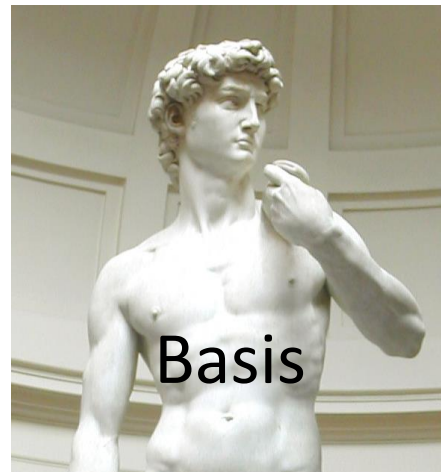
Generation
set



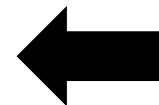
刪去



Same size



疊加



Independent vector set

More from Theorems

Any two bases for a subspace contain the same number of vectors.

$\mathcal{R}^m$ have a basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m\}$ All bases have m vectors

A basis is the smallest generation set.

$$\dim \mathcal{R}^m = m$$

A vector set generates $\mathcal{R}^m$ must contain at least m vectors.

Because a basis is the smallest generation set

Any other generation set has at least m vectors.

A basis is the largest independent set in the subspace.

Any independent vector set in $\mathcal{R}^m$ contain at most m vectors.

Is it a basis?

- Definition: A **basis** B for V is an independent generation set of V.

$$V = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathcal{R}^3 : v_1 - v_2 + 2v_3 = 0 \right\} \quad \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Is $\mathcal{C}$ a basis of V?

Independent? **yes**

Generation set? **difficult**

$$\mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\} \text{ generates } V$$

Is it a basis?

Find a basis for V

- Given a subspace V , assume that we already know that $\dim V = k$.
Suppose S is a subset of V with k vectors

If S is independent $\Rightarrow$ S is basis

If S is a generation set $\Rightarrow$ S is basis

$$V = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathcal{R}^3 : v_1 - v_2 + 2v_3 = 0 \right\} \quad \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$\dim V = 2$ (parametric representation)

Is $\mathcal{C}$ a basis of V ?

$\mathcal{C}$ is a subset of V with 2 vectors
Independent? yes $\Rightarrow$ $\mathcal{C}$ is a basis of V

- Given a subspace V , assume that we already know that $\dim V = k$.
Suppose S is a subset of V with k vectors

If S is independent $\longrightarrow$ S is basis

If S is a generation set $\longrightarrow$ S is basis

5. (8%) Let $A = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a basis for a k -dimensional subspace V of $\mathcal{R}^n$. Prove that $B = \{\mathbf{v}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_k\}$ is also a basis for V , where $\mathbf{v}_1 = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3 + \dots + \mathbf{u}_k$. (2020)

$$c_1 \mathbf{v}_1 + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k = \mathbf{0}$$

$$c_1 (\mathbf{u}_1 + \mathbf{u}_2 + \dots + \mathbf{u}_k) + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k = \mathbf{0}$$

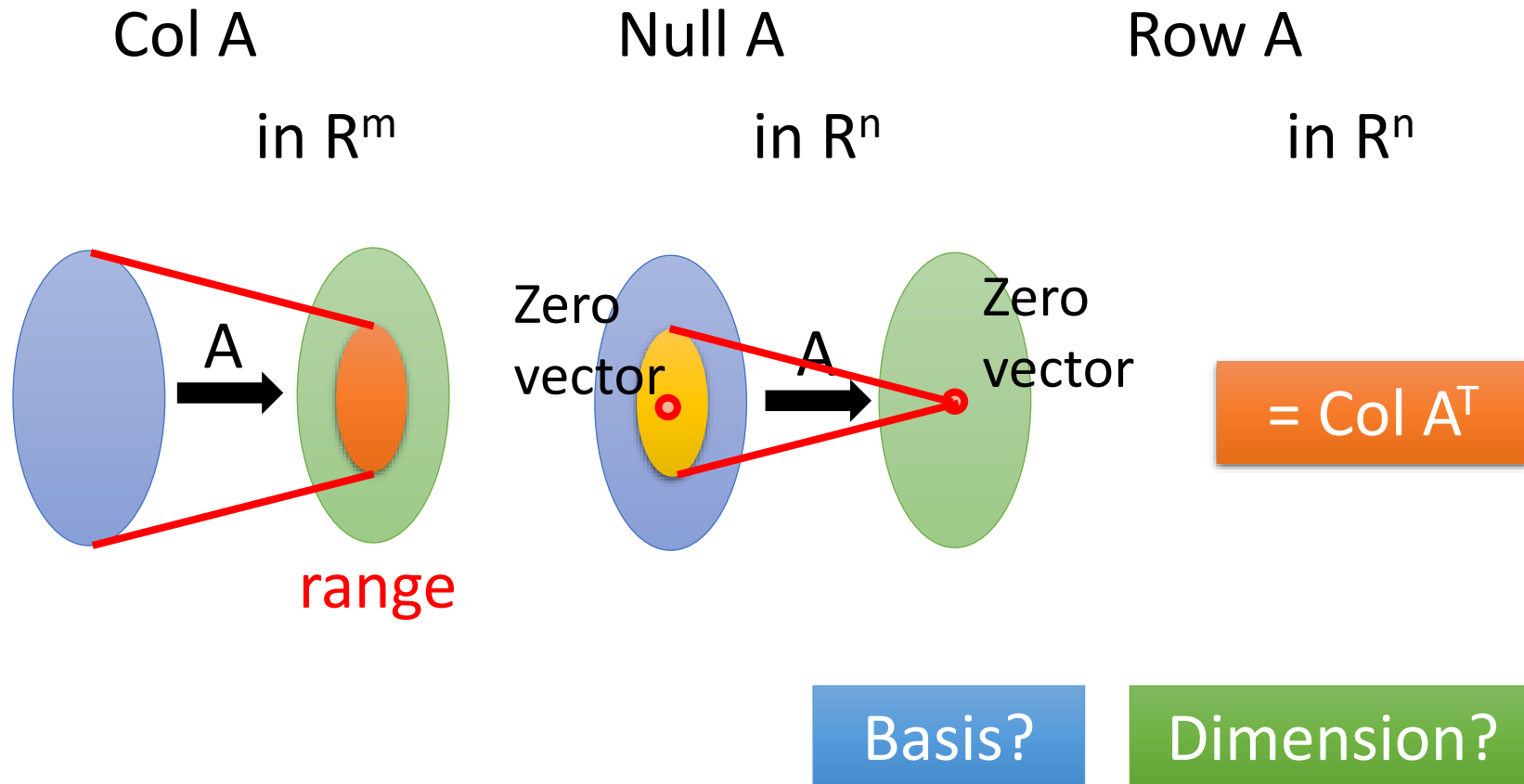
$$c_1 \mathbf{u}_1 + (c_1 + c_2) \mathbf{u}_2 + \dots + (c_1 + c_k) \mathbf{u}_k = \mathbf{0}$$

$$= 0 \qquad = 0 \qquad = 0$$

$$c_1 = c_2 = \dots c_k = 0$$

Three Associated Subspaces

- A is an $m \times n$ matrix



Three Associated Subspaces

- Col A (range of A), $\dim(\text{Col A}) = \text{Rank}(A)$
 - Basis: Finding pivot columns of A
- Null A, $\dim(\text{Null A}) = \text{Nullity}(A)$
 - Basis: Solving $Ax = \mathbf{0}$
- Row A = Col A^T , $\dim(\text{Row A}) = \text{Rank A}$
 - Basis: The nonzero rows of the RREF of A

Original Matrix A v.s. its RREF R

	Relation	Span	Subspace
Column	Unchanged	Changed	$\text{Col A} \neq \text{Col R}$
Row	Changed	Unchanged	$\text{Row A} = \text{Row R}$

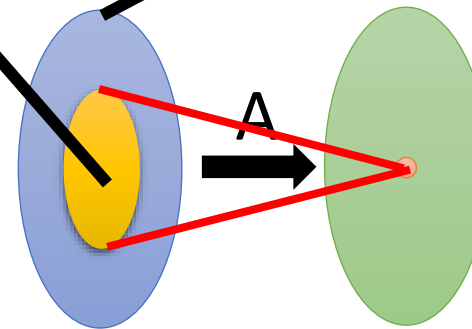
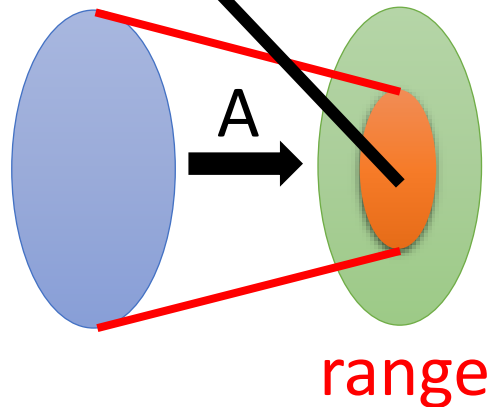
Dimension Theorem

Dim (Col A)
= Rank A

Dim (Null A)
= $n - \text{Rank A}$

If A is $m \times n$
Dim ($\mathbb{R}^n$) = n

$$\text{Dim of Range} + \text{Dim of Null} = \text{Dim of Domain}$$



Summary

A is an $m \times n$ matrix

	Dimension	Basis
Col A	Rank A	The pivot columns of A
Null A	Nullity A $= n - \text{Rank A}$	The vectors in the parametric representation of the solution of $Ax=0$
Row A	Rank A	The nonzero rows of the RREF of A

$$\text{Rank } A = \text{Rank } A^T$$

- Proof

Rank A

= Dim (Col A)

Rank A

= Dim (Row A)

= Dim (Col A^T)

= Rank A^T

Dimension & Basis

2. (10%) Let $T : \mathcal{R}^n \rightarrow \mathcal{R}^m$ be a linear transformation defined by $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathcal{R}^n$, where A is an $m \times n$ matrix. Let $U : \mathcal{R}^m \rightarrow \mathcal{R}^n$ be a linear transformation defined by $U(\mathbf{x}) = A^T \mathbf{x}$ for all $\mathbf{x} \in \mathcal{R}^n$, where A^T is the transpose of A . Prove or disprove that the dimension of the range of T is the same as the dimension of the range of U . (2015)

$$\begin{aligned} T(x) = Ax &\implies \dim \text{ of Col } A = \text{rank of } A \\ U(x) = A^T x &\implies \dim \text{ of Col } A^T = \text{rank of } A^T \end{aligned} \quad \parallel$$

3. Answer whether the following statements are true or false. Justify your answer.
 (a) (5%) The mapping $\phi : R^n \rightarrow R^n$ defined by $\phi(\mathbf{x}) = [\mathbf{x}]_\alpha$ is a linear transformation, where $[\mathbf{x}]_\alpha$ is the coordinate vector of $\mathbf{x}$ relative to a basis α of R^n . (2016)

$$\begin{array}{ccc} & [x]_{\mathcal{B}} = B^{-1}x & \\ x & \xrightarrow{\hspace{10em}} & [x]_{\mathcal{B}} \\ & \xleftarrow{\hspace{10em}} & \\ & x = B[x]_{\mathcal{B}} & \end{array}$$

3. (14%) Let $A = \begin{bmatrix} 1 & 2 & -1 & 2 & 1 \\ -1 & -2 & 1 & 2 & 3 \\ 2 & 4 & -3 & 2 & 0 \\ -3 & -6 & 2 & 0 & 3 \end{bmatrix}.$

(2019)

$$\text{Rank } A = 3, \text{Nullity } A = 2$$

(a) (8%) Find the reduced row echelon form of A .

(b) (6%) Let $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_l\}$ be a basis of $\text{Null } A$. Let $B = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_l]$. Find the dimension of $\text{Null } B^T$.
 2×5 $l = 2$ 5×2

$$\dim(\text{Null } B^T) = \text{Nullity}(B^T)$$

$$= 5 - \text{Rank}(B^T)$$

$$= 5 - \text{Rank}(B) = 5 - \text{Nullity } A = 3$$

(2019)

8. (15%) Let A be an $m \times n$ matrix, and let E_1 and E_2 be an $m \times m$ elementary matrix and an $n \times n$ elementary matrix, respectively. Determine the validity of each of the following statements. You don't have to give a proof if you determine a statement is true; however, if you think a statement is false, find a counterexample to show it (i.e., a specific example of matrices A and E_1 (or E_2)).

- (a) (3%) $\text{Col } A = \text{Col } E_1 A$.  
- (b) (3%) $\text{Row } A = \text{Row } E_1 A$. 
- (c) (3%) $\text{Col } A = \text{Col } AE_2$. 
- (d) (3%) $\text{Null } A = \text{Null } AE_2$.  
- (e) (3%) $\text{Null } A = \text{Null } E_1 A$. 

“Elementary column operation”

$$AE = (E^T A^T)^T$$

Elementary
row operation

Rows of A^T are
columns of A

常見題型: Rank v.s. Matrix Multiplication

Rank A (revisit)

Maximum number of Independent Columns

Number of Pivot Column

Number of Non-zero rows

Number of Basic Variables

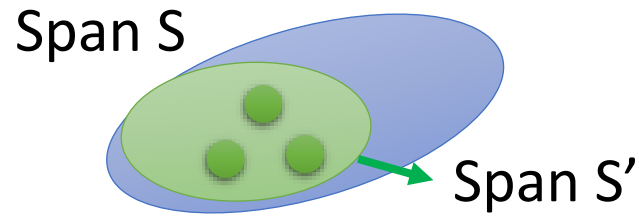
Dim (Col A): dimension of column space

= Dim (Row A)

Dimension of the range of A

= Dim (Col A^T)

Rank



$$\text{Rank } A = \text{Rank } A^T$$

3. Let A and B be $m \times n$ and $n \times k$ matrices respectively. (2014)

(a) (4 + 2%) Prove that $\text{Col } AB$ is a subset of $\text{Col } A$. Also prove that $\text{rank } A \geq \text{rank } AB$.

$$\text{Columns of } AB \subset \text{Col } A \Rightarrow \text{Col } AB \subseteq \text{Col } A$$

$$\Rightarrow \dim(\text{Col } AB) \leq \dim(\text{Col } A) \Rightarrow \text{Rank } AB \leq \text{Rank } A$$

(b) (4%) Suppose that $k = n$ and B is an invertible square matrix. Prove that $\text{Col } A$ is a subset of $\text{Col } AB$.

$$\text{Col } A = \text{Col } ABB^{-1} \subseteq \text{Col } AB$$

$$\begin{aligned} \text{Col } A &= \text{Col } AB \\ \text{Rank } A &= \text{Rank } AB \end{aligned}$$

(d) Show that $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$ for any two matrices. (5%)

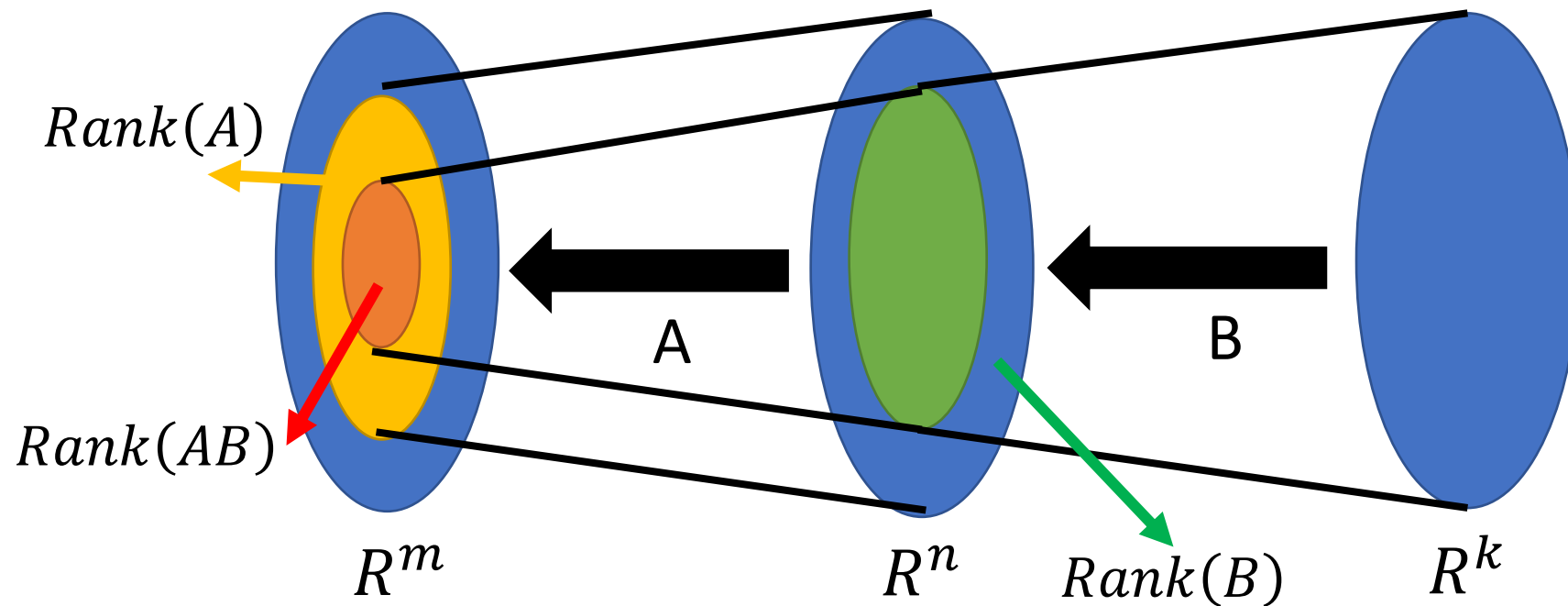
$$\text{Rank}(AB) = \text{Rank}(B^T A^T) \leq \text{Rank}(B^T) = \text{Rank}(B)$$

(b) (4%) Suppose that $k = n$ and B is an invertible square matrix. Prove that $\text{Col } A$ is a subset of $\text{Col } AB$.

Rank – Intuitive Explanation

- If A is a $m \times n$ matrix, and B is a $n \times k$ matrix.

$$\text{Rank}(AB) \leq \min(\text{Rank}(A), \text{Rank}(B))$$

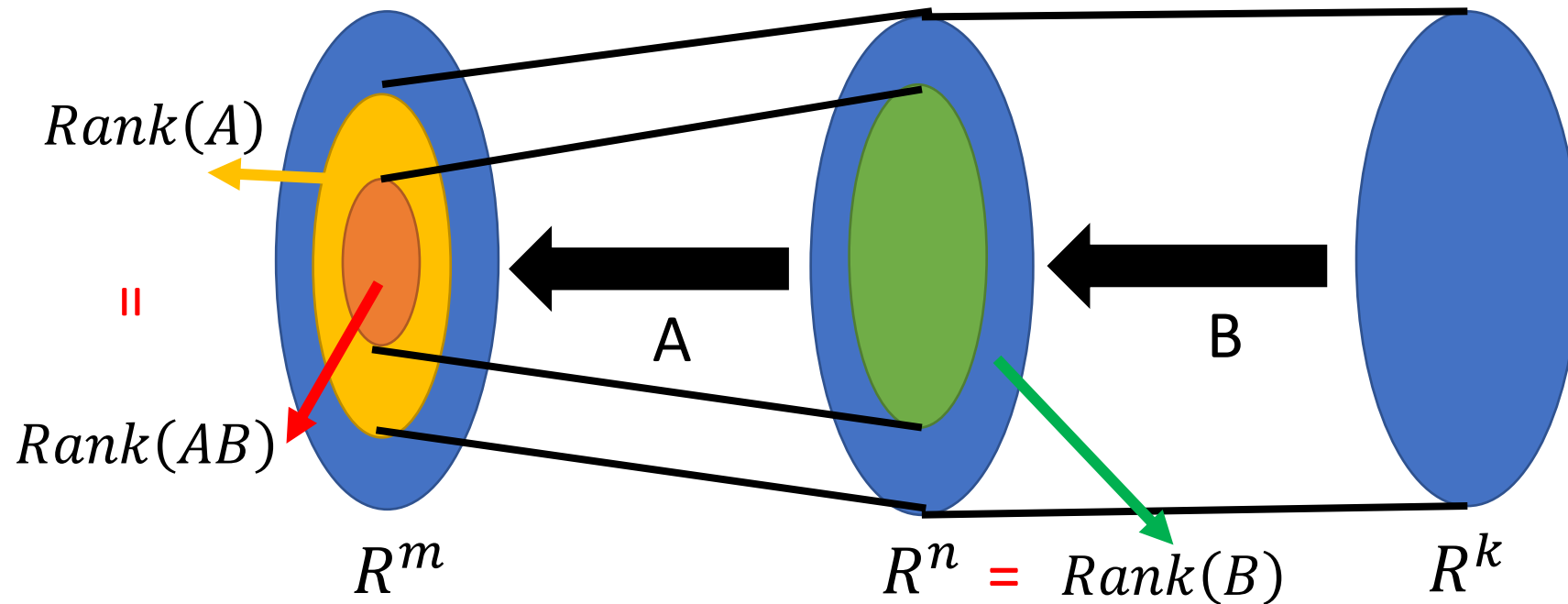


(b) (4%) Suppose that $k = n$ and B is an invertible square matrix. Prove that $\text{Col } A$ is a subset of $\text{Col } AB$.

Rank – Intuitive Explanation

- If A is a $m \times n$ matrix, and B is a $n \times k$ matrix.

$$\text{Rank}(AB) \leq \min(\text{Rank}(A), \text{Rank}(B))$$



(7%) Prove that, for any $m \times n$ matrix A and any $n \times p$ matrix B , the column space of AB is contained in the column space of A .

(2021)

Columns of $AB \subset \text{Col } A \Rightarrow \text{Col } AB \subseteq \text{Col } A$

(7%) Consider an $m \times m$ matrix P and an $m \times n$ matrix A . Which of the following statement is correct?

(2021)

(i) $\text{Null } PA$ is a subspace of $\text{Null } A$; $P = O$

(ii) $\text{Null } A$ is a subspace of $\text{Null } PA$;

(iii) $\text{Null } PA$ may not be a subspace of $\text{Null } A$ and $\text{Null } A$ may not be a subspace of $\text{Null } PA$.

Justify your answer.

Problem 6

(7%) Consider an $m \times n$ matrix A and an $n \times n$ matrix Q . Which of the following statement is correct?

(i) $\text{Null } AQ$ is a subspace of $\text{Null } A$; $Q = O$

(2021)

(ii) $\text{Null } A$ is a subspace of $\text{Null } AQ$;

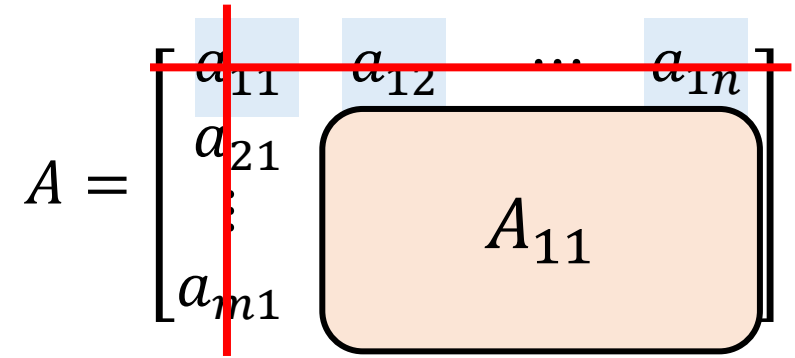
(iii) $\text{Null } AQ$ may not be a subspace of $\text{Null } A$ and $\text{Null } A$ may not be a subspace of $\text{Null } AQ$.

Justify your answer.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad Q = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

常見題型: Determinant

Cofactor Expansion



$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & & & \\ \vdots & & & \\ a_{n1} & & & \end{bmatrix}$$

A_{11}

- Pick row 1

$$\det A = a_{11}c_{11} + a_{12}c_{12} + \cdots + a_{1n}c_{1n}$$

- Or pick row i

$$\det A = a_{i1}c_{i1} + a_{i2}c_{i2} + \cdots + a_{in}c_{in}$$

c_{ij} : (i,j)-cofactor

- Or pick column j

$$\det A = a_{1j}c_{1j} + a_{2j}c_{2j} + \cdots + a_{nj}c_{nj}$$

$$c_{ij} = (-1)^{i+j} \det A_{ij}$$

$$c_{11} = (-1)^{1+1} \det A_{11}$$

Cofactor expansion again ...

Properties of Determinants

- Basic Property 1: $\det(I) = 1$
- Basic Property 2: Exchange rows reverse the sign of det
 - If a matrix A has 2 equal rows, $\det A = 0$
- Basic Property 3: Determinant is “linear” for each row

$$\det \begin{pmatrix} ta & tb \\ c & d \end{pmatrix} = t \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{A row of zeros, } \det A = 0$$

$$\det \begin{pmatrix} a + a' & b + b' \\ c & d \end{pmatrix} = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \det \begin{pmatrix} a' & b' \\ c & d \end{pmatrix}$$

“Subtract k x row i from row j ” does not change det

Determinants for Upper Triangular Matrix

$$U = \begin{bmatrix} d_1 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}$$

Killing everything above
Does not change the det

$$\det(U) = \det \left(\begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix} \right)$$

3-a Property 1

$$= d_1 d_2 \cdots d_n \det \left(\begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} \right)$$

= 1

$$\det(U) = d_1 d_2 \cdots d_n \text{ (Products of diagonal)}$$

(2020)

7. (8%) Calculate $\det$

$$\begin{bmatrix} 0 & 0 & 0 & 4 & 5 \\ 0 & 1 & 2 & 4 & 6 \\ 1 & 2 & 3 & 4 & 5 \\ 0 & 2 & 3 & 4 & 5 \\ 2 & 4 & 6 & 8 & 9 \end{bmatrix}$$

$$\begin{aligned} \det \begin{vmatrix} 0 & 0 & 0 & 4 & 5 \\ 0 & 1 & 2 & 4 & 6 \\ 1 & 2 & 3 & 4 & 5 \\ 0 & 2 & 3 & 4 & 5 \\ 2 & 4 & 6 & 8 & 9 \end{vmatrix} &= \det \begin{vmatrix} 0 & 0 & 0 & 4 & 5 \\ 0 & 1 & 2 & 4 & 6 \\ 1 & 2 & 3 & 4 & 5 \\ 0 & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & -1 \end{vmatrix} \\ &= - \det \begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 4 & 5 \\ 0 & 0 & -1 & -4 & 7 \\ 0 & 0 & 0 & 0 & -1 \end{vmatrix} = \det \begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & -1 & -4 & -7 \\ 0 & 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 0 & -1 \end{vmatrix} \\ &= 1 * 1 * (-1) * 4 * (-1) \end{aligned}$$

(7%) Compute $\det A$, where

(2021)

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 & 1 \\ 1 & 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 & 1 \\ 1 & 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

More Properties of Determinants

$\det E = \det E^T$ in the textbook

- $\det A = \det A^T$
- Proof:

$\det(A) = \text{sum of } n! \text{ terms}$

Format of each term: $a_{\underline{1}\alpha}a_{\underline{2}\beta}a_{\underline{3}\gamma}\cdots a_{\underline{n}\omega}$

Sorted by
column indices



Find an element in
each row

permutation of
 $1, 2, \dots, n$

Format of each term: $a_{\underline{\alpha}'\underline{1}}a_{\underline{\beta}'\underline{2}}a_{\underline{\gamma}'\underline{3}}\cdots a_{\underline{\omega}'\underline{n}}$

Find an element in
each column

permutation of
 $1, 2, \dots, n$

More Properties of Determinants

- $\det(AB) = \det(A)\det(B)$

$$\det(A + B) \neq \det(A) + \det(B)$$

Q: find $\det(A^{-1})$

$$\because A^{-1}A = I \quad \therefore \det(A^{-1})\det(A) = \det(I) = 1$$

$$\therefore \det(A^{-1}) = 1/\det(A)$$

Q: find $\det(A^2)$

$$\det(A^2) = \det(A)\det(A) = \det(A)^2$$

- $\det(A^T) = \det(A)$

- Zero row $\rightarrow$ zero column
- Same row $\rightarrow$ same column

1. (10%) Let A be an $n \times n$ matrix. Let R be the reduced row echelon form of A . Prove or disprove that $\det(A) = \det(R)$. (2015)

Determinant v.s. Invertible

A is invertible



$\det(A) \neq 0$

A



R

Elementary row operation

$\det(A)$

$\det(R)$

$$= \pm k_1 k_2 \cdots \det(A)$$

Exchange: Change sign

If A is invertible, R is identity

$$\det(R) = 1 \Rightarrow \det(A) \neq 0$$

Scaling: Multiply k

If A is not invertible, R has zero row

Add row: nothing

$$\det(R) = 0 \Rightarrow \det(A) = 0$$

Determinant

4. (6%) Let $A = \begin{bmatrix} 1 & 2c & 1-c \\ 0 & 3 & -1-c \\ 0 & -1-c & 3 \end{bmatrix}$. Calculate $\det A$ in terms of c . Determine the conditions on c so that A is invertible. (2018)

5. (5%) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ and let $B = A - cI_3$. Calculate $\det B$ in terms of c . Determine the conditions on c so that B is invertible. (2012)

- How to compute $\det(A)$ (3x3 is usually good enough)
- A is invertible $\iff \det(A) \neq 0$

▼ 祝大家期中考順利!